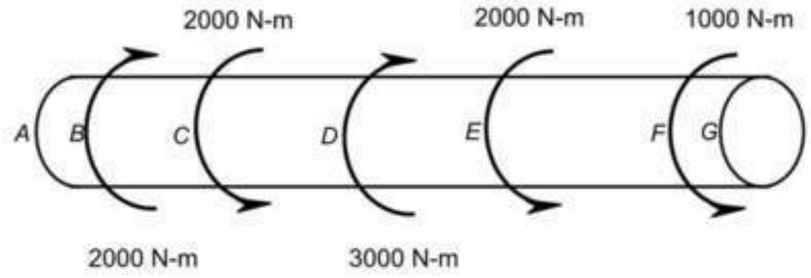
1.

A cylinder is subject to applied torques as shown. In which segment(s) of this cylinder is the torque a maximum?



A. A-B

B. B-C

C. C-D

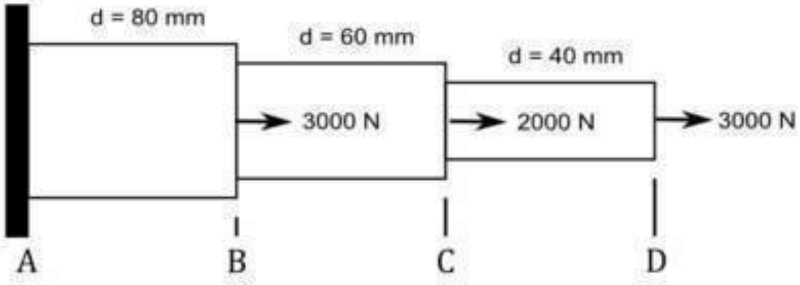
D. D-E

E. E-F

F. F-G

2.

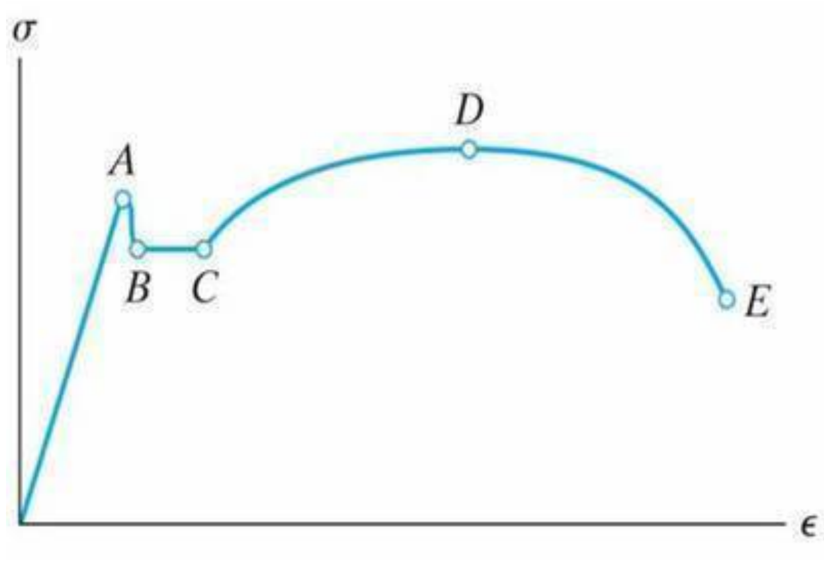
Stepped cylinder A-B-C-D has diameters and is axially loaded as shown. In units of MPa, what is the maximum normal stress in this stepped cylinder ?



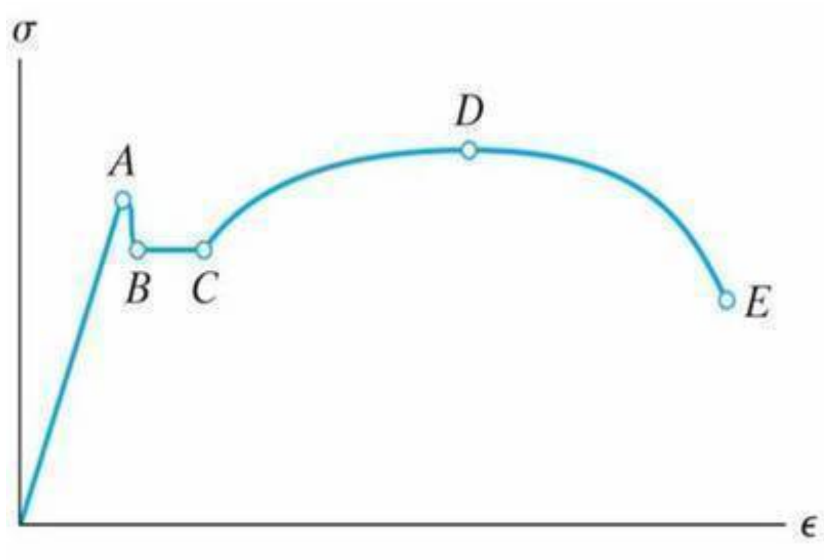
3.

The strain on a member in a truss is measured and found to be ε = 30E-6 . The elongation was also measured and found to be δ = 0.09 mm. In units of meters, what was the original length of this member ?

4.

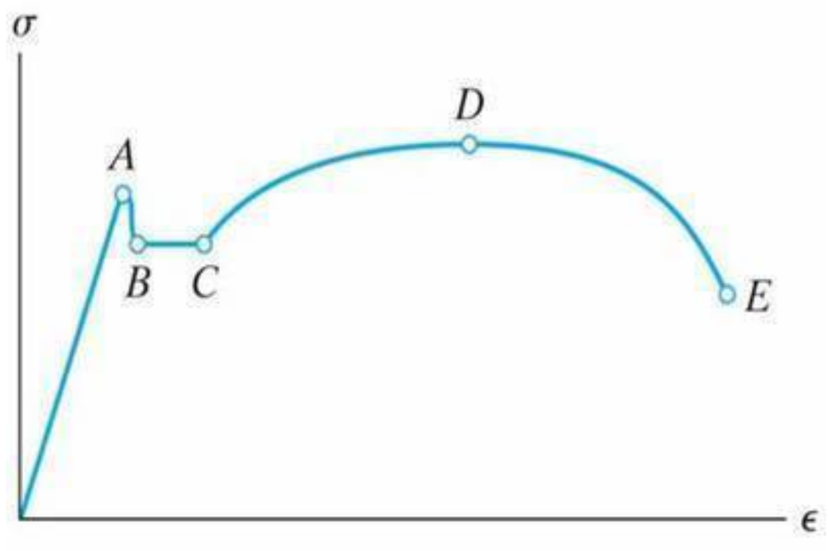


Click or touch the point on this stress-strain diagram corresponding to the fracture stress.



5.

What is represented by point A on the stress strain curve shown ?



A. Ultimate Stress

B. End of Elastic Region

C. Fracture Stress

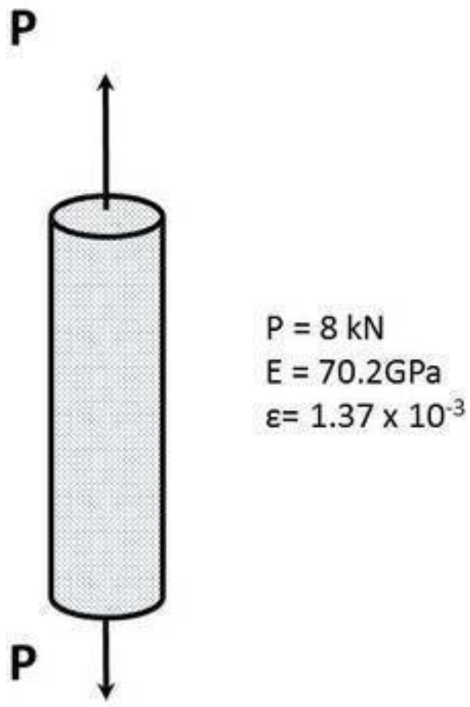
D. Start of Necking

E. Start of Strain Hardening

6.

Consider a cylinder subject to axial loading. It is made out of aluminum (E = 70.2 Gpa) and when a load P = 8kN is applied, a strain gage mounted on its side registers an axial strain ε = 1.37 x 10-3.

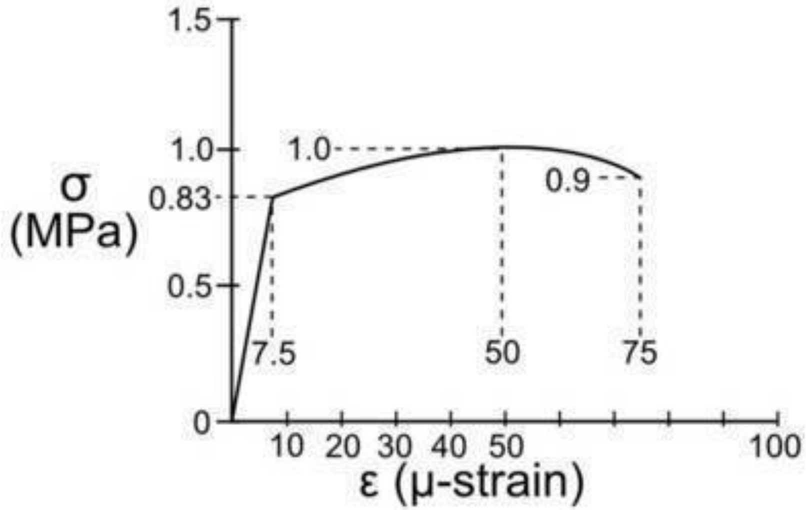
In units of MPa, what is the average normal stress in the cylinder?



7.

A material sample was tested and found to have the stress-strain curve shown. A part to be designed with this material should have a factor of safety F.S.= 6 (based on yield strength).

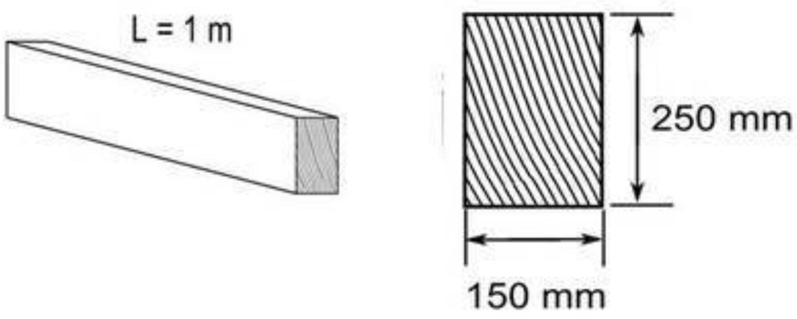
In units of MPa, what is the allowable stress (i.e. design stress) that should be used?



8.

A 1 m long uniform rectangular bar has a cross-section and properties as shown. When the bar is subject to a particular tensile load, the 250 mm dimension shrinks by 0.001mm.

By how much (in mm) did the length increase?



L = 1.0 m
E = 200 Gpa
v = 0.30
w = 0.150 m
H = 0.250 m

9.

A bar is subjectd to an axial tensile stress of $\sigma = 100$ MPa. What is the normal stress on a plane inclined at $\theta = 45^{\circ}$ to the axis of loading

A. 100 MPa

B. 50 MPa

C. 70.7 MPa

D. 25 MPa

10.

A steel component is subjected to an actual working stress of 80 MPa. If the yield stress of the material is 320 MPa, what is the Factor of Safety (FS) based on yield?

A. 2

B. 4

C. 3

D. 6

11.

In mechanical design of ductile materials, which definition of Factor of Safety is most commonly used and why?

A. FS= UTS/Allowed stress, because UTS is always higher

B. FS= Yield stress/ Allowed stress, because ductile materials fail by yielding, not fracture

C. FS = Allowed stress/ yield stress , because we want to stay below yield.

12.

A flat metal bar of rectangular cross section is subjected to a tensile force of 6000N. The bar has a width of 20 mm and a thickness of 5 mm, and is loaded axially (along its length).

Consider a plane inclined 30˙ to the normal plane (i.e., the plane perpendicular to the loading axis). This is equivalent to 60˙ to the loading axis.

Calculate the shear stress τ in MPa on the inclined plane.